1.3 The Exponentials e^t and e^{at}

Here is the key message from this section : **The solutions to $dy/dt = ay$ are $y(t) = Ce^{at}$. That free constant C matches the starting value $y(0)$. Then $y(t) = y(0)e^{at}$.**

I realize that you already know the function $y = e^t$. It is the star of precalculus and calculus. Now it becomes the key to linear differential equations. Here I focus on the two most important properties of this function e^t :

1. *The slope dy/dt equals the function y.* As y grows, its graph gets steeper :

$$\frac{d}{dt}\, e^t = e^t. \tag{1}$$

2. $y(t) = e^t$ follows the *addition rule* for exponents :

$$e^t \text{ times } e^T \text{ equals } e^{t+T}. \tag{2}$$

How is this exponential function constructed? Only calculus can do it, because somewhere we must have a "limit step." Functions from ordinary algebra can get close to e^t, but they can't reach it. If we choose those functions to come closer and closer, then their limit is e^t.

This is like using fractions to approach the extraordinary number π. The fractions can start with 3/1 and 31/10 and 314/100. The neat fraction 22/7 is close to π. But "taking the limit" can't be avoided, because π itself is not a fraction.

Similarly e is not a fraction. On this book's home page **math.mit.edu/dela** is an article called *Introducing e^x*. It describes four popular ways to construct this function. The one chosen now is my favorite, because it is the most direct way.

Construct $\mathbf{y = e^t}$ so that $\dfrac{\mathbf{dy}}{\mathbf{dt}} = \mathbf{y}$ (starting from $y = 1$ at $t = 0$)

To show how this construction works, here are ordinary polynomials y and dy/dt :

1. $y = 1 + t + \dfrac{1}{2}t^2$ The derivative is $dy/dt = 0 + 1 + t$
2. $y = 1 + t + \dfrac{1}{2}t^2 + \dfrac{1}{6}t^3$ The derivative is $dy/dt = 0 + 1 + t + \dfrac{1}{2}t^2$

You see that dy/dt does not fully agree with y. It always falls one term short of y. We could get $t^3/6$ into the derivative by including $t^4/24$ in y. But now dy/dt will be missing $t^4/24$.

You can see that dy/dt won't catch up to y. *The way out is to have infinitely many terms* : *Don't stop*. Then you get $dy/dt = y$.

The limit step reaches an infinite series, adding new terms and never stopping. Every term has the form t^n divided by $n!$ (n *factorial*). Its derivative is the previous term:

$$\textbf{The derivative of } \frac{t^n}{(n)\dots(1)} = \frac{t^n}{n!} \text{ is } \frac{t^{n-1}}{(n-1)\dots(1)} = \frac{t^{n-1}}{(n-1)!} \tag{3}$$

So if $t^n/n!$ is missing in dy/dt, we will capture it by including $t^{n+1}/(n+1)!$ in y.

Of course dy/dt never completely catches up to y—*until we allow an infinite series. There is a term* $t^n/n!$ *for every* n. The term for $n = 0$ is $t^0/0! = 1$.

Construction of e^t

$$y = e^t = 1 + t + \frac{t^2}{2} + \frac{t^3}{6} + \frac{t^4}{24} + \cdots = \sum_{n=0}^{\infty} \frac{t^n}{n!} \tag{4}$$

Taking the derivative of every term produces all the same terms. So $dy/dt = y$. Notice: If you change every t to at, **the derivative of $y = e^{at}$ becomes a times e^{at}:**

$$\frac{d}{dt}\left(1 + at + \frac{a^2t^2}{2} + \frac{a^3t^3}{6} + \cdots\right) = a\left(1 + at + \frac{a^2t^2}{2} + \cdots\right) = ae^{at} \tag{5}$$

This construction of e^t brings up two questions, to be discussed in the Chapter 1 Notes. Does the infinite series add to a finite number (a different number for each choice of t)? Can we add the derivatives of each $t^n/n!$ and safely get the derivative of the sum e^t? Fortunately both answers are *yes*. The terms get very small, very fast, as n increases. The limiting step is $n \to \infty$, producing the exact e^t.

When $t = 1$, we can watch the terms get small. We *must* do this, because $t = 1$ leads to the all-important number e^1 which is e:

The series for e at $t = 1$

$$e = 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \cdots \approx 2.718$$

The first three terms add to 2.5. The first five terms almost reach 2.71. *We never reach* 2.72. With enough terms you can barely pass 2.71828. It is certain that the total sum e is not a fraction. It never appears in algebra, but it is the key number for calculus.

The Series for e^t is a Taylor Series

The infinite series (4) for e^t is the same as the Taylor series. Section 1.2 went from the tangent line $1 + t$ to the tangent parabola $1 + t + \frac{1}{2}t^2$. The next term will be $\frac{1}{6}t^3$, because that matches the third derivative $y''' = 1$ at $t = 0$. *All derivatives are equal to* 1 *at* $t = 0$, when we start from the basic equation $y' = y$. That equation gives $y'' = y' = y$ and the next derivative gives $y''' = y'' = y' = y$.

Conclusion: $t^n/n!$ has the correct n^{th} derivative (which is 1) at the point $t = 0$. All these terms go into the Taylor series. The result is exactly the exponential series (4).

Multiplying Powers by Adding Exponents

We write 3^2 for 3 times 3. We write e^2 for e times e. The question is, does $e = 2.718\ldots$ times $e = 2.718\ldots$ give the same answer as setting $t = 2$ in the infinite series to get e^2?

The answer is again *yes*. I could say "fortunately yes" but that might suggest a lucky accident. The amazing fact is that Property 1 ($y' = y$ is now confirmed) leads automatically to Property 2. The exponential starts from $y(0) = e^0 = 1$ at time $t = 0$.

Property 2. e^t **times** e^T **equals** e^{t+T} so $(e^1)\,(e^1) = e^2$

This is a differential equations course, so the proofs will use Property 1: $dy/dt = y$.

First Proof. We can solve $y' = (a+b)y$ two ways, starting from $y(0) = 1$. We know that $y(t) = e^{(a+b)t}$. Another solution is $y(t) = e^{at}e^{bt}$, as the product rule shows:

$$\frac{d}{dt}\left(e^{at}e^{bt}\right) = \left(ae^{at}\right)e^{bt} + e^{at}\left(be^{bt}\right) = (a+b)e^{at}e^{bt}. \tag{6}$$

This solution $e^{at}e^{bt}$ also starts at $e^0e^0 = 1$. It must be the same as the first solution $e^{(a+b)t}$. The equation $y' = (a+b)y$ only has one solution. At $t = 1$ this says that $e^{a+b} = e^ae^b$. QED.

Second Proof. Starting with $y = 1$ at $t = 0$, the solution out to time t is e^t. The solution to time $t + T$ is e^{t+T}. The question is, do we also get that answer in two steps?

Starting from $y = 1$ at $t = 0$, we go to e^t. Then start from e^t at time t and continue an additional time T. This would give e^T starting from $y = 1$, but here the starting value is e^t. So $C = e^t$ multiplies e^T. At time $t + T$ we have perfect agreement:

e^t times e^T (which is C times e^T) agrees with one big step e^{t+T}.

Negative Exponents

Remember the example $dy/dt = -y$ with solution $y = e^{-t}$. That exponent $-t$ is negative. The solution decays toward zero. The exponent rule $e^te^T = e^{t+T}$ still holds for negative exponents. In particular e^t *times* e^{-t} *is* $e^{t-t} = e^0 = 1$:

Negative exponents $\dfrac{1}{e^t} = e^{-t}$ and $\dfrac{1}{e} = e^{-1} = 1 - 1 + \dfrac{1}{2} - \dfrac{1}{6} + \dfrac{1}{24} - \cdots$

This number $1/e$ is about .36. The series always succeeds! The graph of $y = e^{-t}$ shows that e^{-t} *stays positive*. It is very small for $t > 32$. Your computer might use 32 bit arithmetic and ignore numbers that are this small.

Why does e^t grow so fast? The slope is y itself. So the slope increases when the function increases. That steep slope makes y increase faster—and then the slope too.

Interest Rates and Difference Equations

There is another approach to e^t and e^{at}, which is not based on an infinite series. (At least, not at the start.) It connects to interest on bank accounts. For e^t the rate is $a = 1 = 100\%$. For e^{at} the differential equation is $dy/dt = ay$ and the interest rate is a.

The different approach is to construct e^t and e^{at} as the limit of compound interest.

$$e^t = \lim_{N \to \infty} \left(1 + \frac{t}{N}\right)^N \qquad e^{at} = \lim_{N \to \infty} \left(1 + \frac{at}{N}\right)^N . \tag{7}$$

The beauty of these formulas is that a bank does exactly what a computational scientist does. They both start with the differential equation $dy/dt = ay$ and the initial condition $y = 1$ at $t = 0$. Banks and scientists don't have computers that give exact solutions, when $y(t)$ changes continuously with time. Both take finite time steps Δt instead of infinitesimal steps dt. **They reach time t in N steps of size $\Delta t = t/N$.** Their approximations are $Y_1, Y_2, \ldots, Y_N$ with $Y_0 = 1$. Compound interest produces a **difference equation**:

$$\frac{dy}{dt} = ay \quad \text{becomes} \quad \frac{Y_{n+1} - Y_n}{\Delta t} = a\,Y_n \quad \text{and} \quad \boldsymbol{Y_{n+1} = (1 + a\,\Delta t)Y_n}. \tag{8}$$

Each step multiplies the bank balance by $1 + a\Delta t$. The new balance is the old balance Y_n plus $a\,\Delta t Y_n$ (the interest on Y_n in the time interval Δt). This is ordinary compound interest that all banks offer, not continuous compounding as in dy/dt. The time step can be $\Delta t = 1$ year or 1 month. The balance at $t = 2$ years $= 24$ months is Y_2 or Y_{24}:

$$Y_2 = (1 + a)^2\, Y_0 \qquad Y_{24} = \left(1 + \frac{a}{12}\right)^{24} Y_0 \approx e^{2a} Y_0. \tag{9}$$

If the rate is $a = 3$ per cent per year $= .03$ per year, continuous compounding for 2 years would produce the exponential factor $e^{.06} \approx 1.06184$. Monthly compounding produces $(1.0025)^{24} \approx 1.06176$. We only lose a little, when the differential equation $y' = ay$ is approximated by the difference equation in (8).

The computational scientist is usually not willing to accept this loss of accuracy in Y. Equation (8) with a forward difference $Y_{n+1} - Y_n$ is called **Euler's method**. Its accuracy is not high and not hard to improve. It is the natural choice for a bank, because a backward difference costs them even more than continuous compounding:

$$\textbf{Backward difference} \qquad \frac{Y_n - Y_{n-1}}{\Delta t} = aY_n \quad \text{or} \quad Y_n = \frac{1}{1 - a\Delta t} Y_{n-1}. \tag{10}$$

Y_n connects backward to the earlier Y_{n-1}. Now each step divides by $1 - a\Delta t$. After N steps of size $\Delta t = t/N$, we are again close to e^{at}. But with backward differences and $a > 0$, we overshoot the differential equation and the bank pays a little too much:

$$(1 + a\Delta t)^N \text{ is below } e^{at} \qquad \frac{1}{(1 - a\Delta t)^N} \textit{ is above } e^{at}.$$

Complex Exponents

This isn't the time and place to study complex numbers in detail. It will be the pages about oscillations and $e^{i\omega t}$ that cannot go forward without the imaginary number i. Here we are solving $dy/dt = ay$, and all I want to do is to **choose $a = i$.**

I can think of two ways to solve the complex equation $\boldsymbol{dy/dt = iy}$. The fast way uses derivatives of sine and cosine, which we know well:

$$\begin{aligned} &\textbf{Proposed solution} && y = \cos t + i \sin t \\ &\textbf{Compare } \boldsymbol{dy/dt} && dy/dt = -\sin t + i \cos t \\ &\textbf{with the right side } \boldsymbol{iy} && iy = i \cos t + i^2 \sin t \end{aligned} \tag{11}$$

To check $dy/dt = iy$, compare the last two lines. **Use the rule $\mathbf{i^2 = -1}$.** (We had to imagine this number, because no real number has $x^2 = -1$.) Then $-\sin t$ is the same as $i^2 \sin t$. So $\boldsymbol{y = \cos t + i \sin t}$ solves the equation $dy/dt = iy$. This solution starts at $y = 1$ when $t = 0$, because $\cos 0 = 1$ and $\sin 0 = 0$.

The slower approach to $dy/dt = iy$ uses the infinite series. Since $a = i$, the solution e^{at} becomes e^{it}. Formally, the series for $y = e^{it}$ certainly solves $dy/dt = iy$:

$$\textbf{Complex exponential} \qquad \boldsymbol{y = e^{it}} = 1 + (it) + \frac{1}{2}(it)^2 + \frac{1}{6}(it)^3 + \cdots \tag{12}$$

The derivative of each term is i times the previous term. Since the series never stops, the derivative dy/dt perfectly matches iy. And we are still starting at $y = 1$ when we substitute $t = 0$. **This infinite series $\boldsymbol{e^{it}}$ equals the first solution $\boldsymbol{\cos t + i \sin t}$.**

Now use the rule $i^2 = -1$. For $(it)^2$ I will write $-t^2$. And $(it)^3$ equals $-it^3$. The fourth power of i is $i^4 = i^2 i^2 = (-1)^2 = 1$. That sequence $\boldsymbol{i, -1, -i, 1}$ repeats forever.

$$i = i^5 \qquad i^2 = i^6 = -1 \qquad i^3 = i^7 = -i \qquad i^4 = i^8 = 1$$

The infinite series (12) includes those four numbers multiplying powers of t:

$$e^{it} = 1 + \left[it - 1\frac{t^2}{2!} - i\frac{t^3}{3!} + 1\frac{t^4}{4!}\right] + \left[i\frac{t^5}{5!} - 1\frac{t^6}{6!} - i\frac{t^7}{7!} + 1\frac{t^8}{8!}\right] + \cdots$$

This may be the first time a textbook has ever written out nine terms. You can see the full repeat of i, -1, $-i$, 1. That last coefficient divides by $8! = 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$ which is 40320.

The main point is that the solution $y = \cos t + i \sin t$ in equation (11) must be the same as this series solution e^{it}. They both solve $dy/dt = iy$. They both start at $y = 1$ when $t = 0$. The equality between them is one of the greatest formulas in mathematics.

$$\boxed{\textbf{Euler's Formula is} \qquad \boldsymbol{e^{it} = \cos t + i \sin t.}} \tag{13}$$

Then $e^{i\pi} = \cos \pi + i \sin \pi = \mathbf{-1}$. And $e^{i2\pi} = 1 + i2\pi + \frac{1}{2}(i2\pi)^2 + \cdots$ must add to $\mathbf{1}$!

I cannot resist comparing $\cos t + i \sin t$ with the series for e^{it}. The **real part** of that series must be $\cos t$. The **imaginary part** (which multiplies i) must be $\sin t$. The even powers $1, t^2, t^4, \ldots$ give cosines. The odd powers $t, t^3, t^5, \ldots$ are multiplied by i:

Cosine is even
$$\cos t = 1 - \frac{1}{2}t^2 + \frac{1}{24}t^4 - \frac{t^6}{6!} + \cdots \tag{14}$$

Sine is odd
$$\sin t = t - \frac{1}{6}t^3 + \frac{1}{120}t^5 - \frac{t^7}{7!} + \cdots \tag{15}$$

These two pieces of the series for e^{it} are famous functions on their own, and now we see their Taylor series . They are beautifully connected by Euler's Formula.

The derivative of the sine series is the cosine series :

$$\frac{\mathbf{d}}{\mathbf{dt}}\mathbf{\sin t = \cos t} \qquad \frac{d}{dt}\left(t - \frac{1}{6}t^3 + \cdots\right) = \mathbf{1 - \frac{1}{2}t^2 + \cdots = \text{cosine}}$$

The derivative of the cosine series is minus the sine series :

$$\frac{\mathbf{d}}{\mathbf{dt}}\mathbf{\cos t = -\sin t} \qquad \frac{d}{dt}\left(1 - \frac{1}{2}t^2 + \frac{1}{24}t^4 - \cdots\right) = \mathbf{-t + \frac{1}{6}t^3 \cdots = -\,\text{sine}}$$

All this important information came from allowing the exponent in e^{it} to be imaginary. And e^{it} times e^{-it} is exactly $\cos^2 t + \sin^2 t = 1$.

Matrix Exponents

One more thing, which you can safely ignore for now. The exponent in e^{at} could become a **square matrix**. Instead of solving $dy/dt = ay$ by e^{at}, we can solve the matrix equation $d\boldsymbol{y}/dt = A\boldsymbol{y}$ by the matrix e^{At}. Start with the identity matrix I instead of the number 1.

e^{At} is a matrix
$$e^{\boldsymbol{At}} = \boldsymbol{I + At + \frac{1}{2}(At)^2 + \frac{1}{6}(At)^3 + \cdots} \tag{16}$$

The series has the usual form, with the matrix A instead of the number a. Here I stop, because matrices come in Chapter 4 : *Systems of Equations*. When the matrix A is three by three, the equation $dy/dt = Ay$ represents three ordinary differential equations. Still first order linear, still constant coefficients, solved by e^{At} in Section 6.4.

There is one big difference for matrices: $\boldsymbol{e^{At}e^{Bt} = e^{(A+B)t}}$ **is not true**. For numbers a and b this equation is correct. For matrices A and B something goes wrong in equation (6). When you look closely, you see that b moved in front of e^{at}. But $e^{At}B = Be^{At}$ is false for matrices.

■ REVIEW OF THE KEY IDEAS ■

1. In the series for e^t, each term $t^n/n!$ is the derivative of the next term.
2. Then the derivative of e^t is e^t, and the exponent rule holds: $e^t e^T = e^{t+T}$.
3. Another approach to $dy/dt = y$ is by finite differences $(Y_{n+1} - Y_n)/\Delta t = Y_n$. $Y_{n+1} = Y_n + \Delta t Y_n$ is the same as compound interest. Then Y_n is close to $e^{n\Delta t} Y_0$.
4. $y = e^{at}$ solves $y' = ay$, and $a = i$ leads to $e^{it} = \cos t + i \sin t$ (Euler's Formula).
5. $\cos t = 1 - t^2/2 + \cdots$ and $\sin t = t - t^3/6 + \cdots$ are the even and odd parts of e^{it}.

Problem Set 1.3

1 Set $t = 2$ in the infinite series for e^2. The sum must be e times e, close to 7.39. How many terms in the series to reach a sum of 7 ? How many terms to pass 7.3 ?

2 Starting from $y(0) = 1$, find the solution to $dy/dt = y$ at time $t = 1$. Starting from that $y(1)$, solve $dy/dt = -y$ to time $t = 2$. Draw a rough graph of $y(t)$ from $t = 0$ to $t = 2$. What does this say about e^{-1} times e ?

3 Start with $y(0) = \$5000$. If this grows by $dy/dt = .02y$ until $t = 5$ and then jumps to $a = .04$ per year until $t = 10$, what is the account balance at $t = 10$?

4 Change Problem 3 to start with $5000 growing at $dy/dt = .04y$ for the first five years. Then drop to $a = .02$ until $t = 10$. What is now the balance at $t = 10$?

Problems 5–8 are about $y = e^{at}$ and its infinite series.

5 Replace t by at in the exponential series to find e^{at}:

$$e^{at} = 1 + at + \frac{1}{2}(at)^2 + \cdots + \frac{1}{n!}(at)^n + \cdots$$

Take the derivative of every term (keep five terms). Factor out a to show that *the derivative of* e^{at} *equals* ae^{at}. At what time T does e^{at} reach 2 ?

6 Start from $y' = ay$. Take the derivative of that equation. Take the nth derivative. Construct the Taylor series that matches all these derivatives at $t = 0$, starting from $1 + at + \frac{1}{2}(at)^2$. Confirm that this series for $y(t)$ is the series for e^{at} in Problem 5.

7 At what times t do these events happen ?

(a) $e^{at} = e$ (b) $e^{at} = e^2$ (c) $e^{a(t+2)} = e^{at}e^{2a}$.

8 If you multiply the series for e^{at} in Problem 5 by itself you should get the series for e^{2at}. Multiply the first 3 terms by the same 3 terms to see the first 3 terms in e^{2at}.

9 (recommended) Find $y(t)$ if $dy/dt = ay$ and $\boldsymbol{y(T) = 1}$ (instead of $y(0) = 1$).

10 (a) If $dy/dt = (\ln 2)y$, explain why $y(1) = 2y(0)$.

(b) If $dy/dt = -(\ln 2)y$, how is $y(1)$ related to $y(0)$?

11 In a one-year investment of $y(0) = \$100$, suppose the interest rate jumps from 6% to 10% after six months. Does the equivalent rate for a whole year equal 8%, or more than 8%, or less than 8% ?

12 If you invest $y(0) = \$100$ at 4% interest compounded continuously, then $dy/dt = .04y$. Why do you have more than $\$104$ at the end of the year ?

13 What linear differential equation $dy/dt = a(t)y$ is satisfied by $y(t) = e^{\cos t}$?

14 If the interest rate is $a = 0.1$ per year in $y' = ay$, how many years does it take for your investment to be multiplied by e ? How many years to be multiplied by e^2 ?

15 Write the first four terms in the series for $y = e^{t^2}$. Check that $dy/dt = 2ty$.

16 Find the derivative of $Y(t) = (1 + \dfrac{t}{n})^n$. If n is large, this dY/dt is close to Y !

17 Suppose the exponent in $y = e^{u(t)}$ is $u(t) =$ integral of $a(t)$. What equation $dy/dt =$ _____ y does this solve ? If $u(0) = 0$ what is the starting value $y(0)$?

Challenge Problems

18 $e^{d/dx} = 1 + d/dx + \frac{1}{2}(d/dx)^2 + \cdots$ is a sum of higher and higher derivatives. Applying this series to $f(x)$ at $x = 0$ would give $f + f' + \frac{1}{2}f'' + \cdots$ at $x = 0$. The Taylor series says : This is equal to $f(x)$ at $x =$ _____.

19 (Computer or calculator, 2.xx is close enough) Find the time t when $e^t = 10$. The initial $y(0)$ has increased by an order of magnitude––a factor of 10. The exact statement of the answer is $t =$_____. At what time t does e^t reach 100 ?

20 The most important curve in probability is the bell-shaped graph of $e^{-t^2/2}$. With a calculator or computer find this function at $t = -2, -1, 0, 1, 2$. Sketch the graph of $e^{-t^2/2}$ from $t = -\infty$ to $t = \infty$. *It never goes below zero.*

21 Explain why $y_1 = e^{(a+b+c)t}$ is the same as $y_2 = e^{at}e^{bt}e^{ct}$. They both start at $y(0) = 1$. They both solve what differential equation ?

22 For $y' = y$ with $a = 1$, Euler's first step chooses $Y_1 = (1 + \Delta t)Y_0$. Backward Euler chooses $Y_1 = Y_0/(1 - \Delta t)$. Explain why $1 + \Delta t$ is smaller than the exact $e^{\Delta t}$ and $1/(1 - \Delta t)$ is larger than $e^{\Delta t}$. (Compare the series for $1/(1 - x)$ with e^x.)

Note Section 3.5 presents an accurate Runge-Kutta method that captures three more terms of $e^{a\Delta t}$ than Euler. For $dy/dt = ay$ here is the step to Y_{n+1} :

Runge-Kutta for $\boldsymbol{y' = ay}$ $\quad Y_{n+1} = \left(1 + a\Delta t + \dfrac{a^2\Delta t^2}{2} + \dfrac{a^3\Delta t^3}{6} + \dfrac{a^4\Delta t^4}{24}\right)Y_n.$